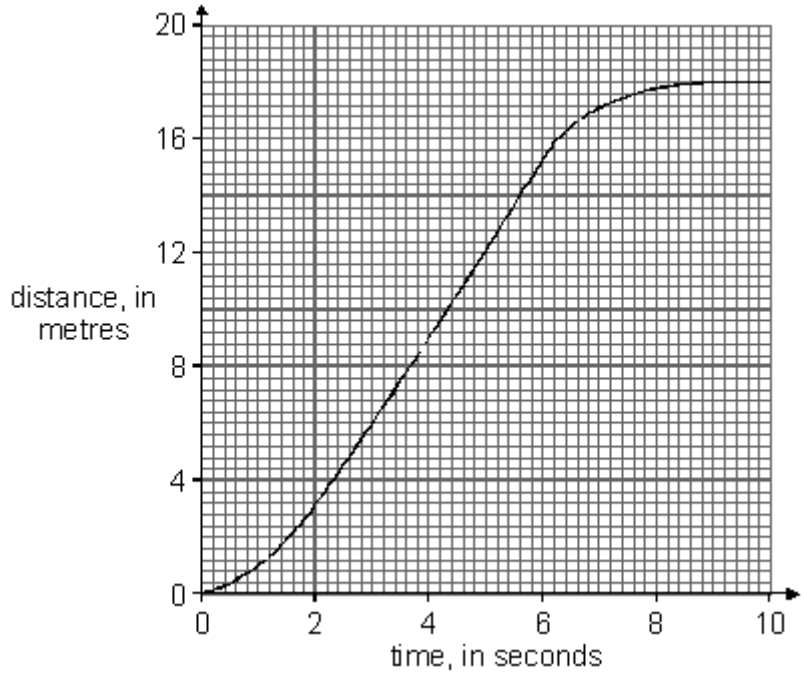
Q1. A remote-controlled car was timed over a period of 10 seconds.
A graph of **distance** against **time** is shown below.



(a) Describe the motion of the car between:

(i) 2 seconds and 6 seconds;

.....

1 mark

(ii) 9 seconds and 10 seconds.

.....

1 mark

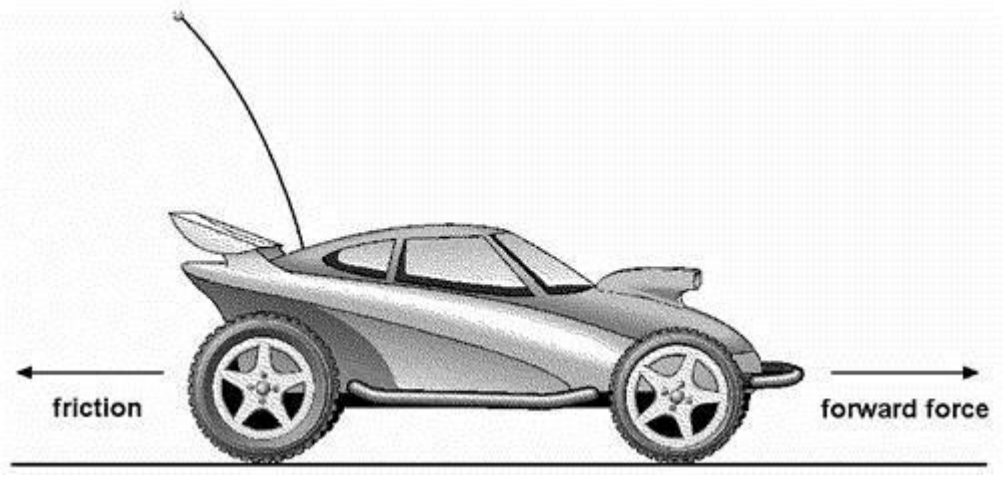
(b) Calculate the average speed of the car between 0 and 10 seconds.
Give the unit.

.....

.....

2 marks

(c) The diagram below shows **two** of the forces acting on the car when it is moving.



(i) When the motor was switched off, the car slowed down and then stopped.

While the car was slowing down, which of the following was true? Tick the correct box.

Friction was zero and the forward force was greater than zero.	☐
The forward force was zero and friction was greater than zero.	☐
Friction was zero and the forward force was zero.	☐
The forward force and friction were both greater than zero.	☐

1 mark

(ii) Use the graph to find the time when the car started to slow down.

The car started to slow down after ..... s.

1 mark

Maximum 6 marks

Q1.

(a) (i) constant speed **or** steady speed

accept not accelerating

1 (L7)

(ii) stationary **or** not moving **or** stopped

accept 'steady speed of zero'

*do **not** accept 'it has a steady speed'*

1 (L7)

(b) 1.8

accept $\frac{18}{10}$

1 (L7)

m/s

*accept 'metres per second' **or** 'ms^{-1}'*

*do **not** accept 'mps'*

1 (L7)

(c) (i) The forward force was zero and friction was greater than zero. ✓

if more than one box is ticked, award no mark

1 (L7)

(ii) 6

accept answers from 5.8 to 6.2

1 (L7)

[6]